

Project Report: IE500 Data Mining

Ethnic Bias and Generalization in Facial Attractiveness Prediction from Geometric Beauty Markers

Team Number: 3

Maxim Froehlich, David Siregar, Timon Kuhl,
Lars Bullmahn, Simon Heinz

May 17, 2026

Submitted to

Prof. Dr. Christian Bizer
University of Mannheim
Data and Web Science Group

Contents

1	Application Area and Goals	1
2	Data Profile	1
2.1	Datasets	1
2.2	Attribute Overview and Label Normalization	2
3	Preprocessing and Feature Engineering	2
3.1	Landmark Extraction and Feature Computation	2
3.2	Data Cleaning	3
4	Data Mining Methods	3
4.1	Models	4
4.2	Hyperparameter Optimization and Evaluation	4
5	Results and Discussion	4
5.1	Overall Model Performance	5
5.2	Decision Tree Analysis	5
5.3	Cross-Dataset Generalization	6
5.4	Fairness Analysis: Multi-Model Bias Comparison	6
5.5	Feature Importance and SHAP Analysis	7
5.6	Comparison to Benchmarks	8
5.6.1	vs. Most-Upvoted Kaggle Notebook	8
5.6.2	vs. Academic SOTA (FPEM)	8
5.7	Limitations	9
6	Conclusion	9
	Bibliography	10
	Ehrenwörtliche Erklärung	11

1 Application Area and Goals

Facial attractiveness prediction is relevant to facial surgery planning, dating platforms that rank user profiles, and social media algorithms suspected of boosting attractive faces. Unlike most computer vision tasks, attractiveness ratings are inherently subjective and culturally shaped: what is considered attractive varies across ethnic groups and geographies. Models trained on data from a narrow demographic may learn culture-specific beauty standards rather than broadly applicable visual patterns, raising both methodological and ethical concerns.

To understand which facial proportions drive attractiveness ratings and whether these patterns vary across ethnicities, we extract 30 geometric ratios and 9 expression features from MediaPipe [1] landmarks and compare several regression approaches. Named features allow direct interpretation of model behavior, unlike pixel-level deep learning models where individual predictions remain opaque.

Our three research questions are:

1. Can geometric facial ratios predict attractiveness from landmarks alone, and how does this compare to deep learning approaches?
2. How does training set composition and diversity affect cross-ethnic generalization?
3. Which geometric measurements drive predictions, and do their weightings differ across ethnic groups?

2 Data Profile

2.1 Datasets

We combine three publicly available facial beauty datasets to create a diverse corpus of 17,870 annotated face images, as shown in table 2.1.

SCUT-FBP5500 [2] contains 5,500 front-facing images (2,750 male, 2,750 female) from Asian and Caucasian subjects, each rated by 60 annotators on a 1–5 Likert scale. **MEBeauty** [3] provides 2,370 images across six ethnic groups (Asian, Black, Caucasian, Hispanic, Indian, Middle Eastern) rated by approximately 300 Amazon Mechanical Turk workers on a 1–10 scale, making it significantly more diverse but smaller.

LiveBeauty [4] contributes 10,000 Asian faces with 200,000 total pairwise beauty labels converted to mean scores on a 1–5 scale. However, its score distribution is near-binary: 45.6% score 3 and 49.8% score 4, with 95.4% of samples in just two bins. This makes it

Table 2.1: Summary of datasets used.

Dataset	N	Scale	Ethnicities	Raters/Image	Source
SCUT-FBP5500	5,500	1–5	2	60	Kaggle
MEBeauty	2,370	1–10	6	~300	GitHub
LiveBeauty	10,000	1–5	1	~20	Tianchi

effectively a classification rather than regression target and inflates aggregate metrics for models trained on it.

After feature extraction via MediaPipe Face Mesh (Chapter 3), the final tabular dataset contains 17,870 rows and 50 columns: 30 geometric beauty markers, 9 expression-related features, and metadata columns (dataset source, ethnicity, gender, raw score, z-normalized score). The extraction success rate was 99.7% (54 images failed).

The combined dataset is heavily imbalanced: Asian faces account for 80.3% (14,351), Caucasian 13.9% (2,480), and remaining groups 5.8% (Black: 296, Hispanic: 296, Middle Eastern: 291, Indian: 156).

2.2 Attribute Overview and Label Normalization

All features are **normalized relative to face dimensions** (ratios to face width or height), making them resolution-invariant. Since the three datasets use different rating scales (1–5, 1–10, 1–5), we apply **z-score normalization per dataset**: $z_i = (x_i - \mu_d) / \sigma_d$. We considered alternatives: (1) **min-max scaling** to $[0,1]$, which is sensitive to outliers; and (2) **percentile-rank transformation**, which discards distance information. We chose z-scores because they preserve relative distances while aligning center and scale, which is important for regression.

3 Preprocessing and Feature Engineering

3.1 Landmark Extraction and Feature Computation

We use Google’s MediaPipe Face Mesh [1] to detect 478 facial landmarks plus 52 blendshape coefficients per image. The pipeline operates in three steps: (1) face detection (single-face constraint), (2) landmark localization (468 facial + 10 iris landmarks), and (3) blendshape estimation (52 expression coefficients). All processing runs on CPU, requiring approximately 0.1 seconds per image.

From the 478 landmarks, we compute 30 geometric beauty markers organized into five categories (Table 3.1). All distances are expressed as **ratios to face width or height**, eliminating dependence on image resolution. Canthal tilt is corrected for head roll using the inter-eye baseline angle, isolating true eye slant from head pose. Bilateral symmetry is computed from 9 paired landmark distances relative to the facial midline, and **phi_deviation** measures how far the face deviates from the golden ratio ($\varphi = 1.618$). Additionally, 9 expression features derived from blendshapes (smile intensity, frown, jaw openness, etc.) control for expression confounds.

Table 3.1: Summary of extracted feature categories (30 geometric features).

Category	Count	Example Features
Eyes	9	canthal_tilt, eye_width_ratio, eye_area_ratio, eye_asymmetry
Nose	3	nose_width_ratio, nose_length_ratio, nose_symmetry
Mouth/Lips	7	lip_fullness_ratio, mouth_chin_ratio, cupids_bow_ratio
Face & Jaw	7	facial_symmetry, jaw_width_ratio, gonial_angle, chin_taper
Proportions	4	midface_ratio, lower_face_ratio, phi_deviation, facial_thirds_symmetry

3.2 Data Cleaning

No missing value imputation was required since MediaPipe either produces a full landmark set or fails entirely (54 images removed). Features with physically implausible values are capped at the 99.5th percentile.

Outlier detection: We flag samples with high annotator disagreement by computing the per-sample standard deviation of rater scores. Samples in the top 10% by standard deviation within each dataset are marked as “high variance.” This information is available for SCUT-FBP5500 and MEBeauty; for LiveBeauty, variance data is largely unavailable ($\approx 90\%$ zero values), so flagging is not applied there. Flagged samples are retained in the analysis.

Attractiveness scores are z-normalized per dataset (mean=0, std=1); predictions can be mapped back via $\hat{x} = \hat{z} \cdot \sigma_d + \mu_d$.

4 Data Mining Methods

4.1 Models

We train seven distinct models spanning different algorithmic families, all taking the same 39-feature input vector (30 geometric + 9 expression). As **baselines**, we use a Global Mean Predictor (always predicts the training-set mean z-score) and **Ridge Regression** (a linear model that separates the contribution of nonlinear interactions from raw linear signal).

We compare three **gradient-boosted tree** variants: **XGBoost** [5] (1,000 trees, max depth 6, learning rate 0.05, early stopping with patience=50), **LightGBM** [6] (leaf-wise growth, 31 leaves, histogram-based splits), and **CatBoost** [7] (ordered boosting with 1,000 iterations). A **Stacking Ensemble** combines four base learners (XGBoost, Random Forest, Gradient Boosting, Ridge) via a Ridge meta-learner, using internal 5-fold CV to generate out-of-fold meta-features. We also train a **3-layer MLP** (128–64–32 neurons, ReLU, Adam optimizer, early stopping on validation loss). A **Quantile XGBoost** variant predicts the median rather than the mean, reducing outlier sensitivity.

4.2 Hyperparameter Optimization and Evaluation

Hyperparameters for gradient-boosted trees are optimized with **Optuna** [8] (TPE sampling, 200 trials) over $\text{max_depth} \in [3, 10]$, $\text{learning_rate} \in [0.01, 0.3]$ (log-uniform), $\text{n_estimators} \in [100, 2000]$, $\text{subsample/colsample_bytree} \in [0.5, 1.0]$, and $\text{L1/L2 regularization} \in [10^{-8}, 10]$, minimizing 5-fold CV MAE.

Holdout Test Set: We use a stratified 80/20 train/test split (stratified by ethnicity) with 5-fold CV on the training portion. Within each fold, 10% is held out for early stopping. **Cross-Dataset Generalization:** We train on one dataset and evaluate on the others (6 combinations) to test ethnic transfer. **Fairness:** Using out-of-fold predictions, we compute per-ethnicity and per-gender metrics. Fairness is quantified via the absolute fairness gap ($\text{max group MAE} - \text{min group MAE}$).

Our primary metric is **MAE** (Mean Absolute Error), reported on each dataset’s native rating scale (1–5 for SCUT and LiveBeauty, 1–10 for MEBeauty). We train on z-normalized scores to combine datasets with different scales, but convert prediction errors back to native units for interpretability. We adopt Spearman ρ as the primary correlation metric because it measures ranking ability independent of target distribution shape, making it valid across both continuous (SCUT, MEBeauty) and near-binary (LiveBeauty) targets. Pearson r is reported alongside for comparability with prior literature.

5 Results and Discussion

5.1 Overall Model Performance

We evaluate all models against a **Global Mean Predictor** baseline on the combined holdout test set (stratified 80/20 split, 3,574 test samples). Table 5.1 shows correlation metrics and MAE improvement over baseline.

Table 5.1: Model performance on held-out test set (combined across all datasets).

Model	Spearman ρ	Pearson r	vs. Baseline ¹
Global Mean (baseline)	—	—	—
Ridge Regression	0.650	0.660	−30.0%
Decision Tree	0.538	0.546	−22.7%
Random Forest	0.648	0.657	−27.8%
XGBoost	0.700	0.708	−35.4%
LightGBM	0.697	0.705	−35.2%
CatBoost	0.701	0.709	−35.3%
Stacking Ensemble	0.702	0.710	−35.4%
MLP (Neural Network)	0.706	0.714	−36.6%
Quantile XGBoost	0.688	0.696	−34.7%

All learned models substantially outperform the baseline, confirming that the 30 geometric features carry genuine predictive signal. Even Ridge Regression reduces MAE by 30%, showing meaningful linear structure. The gradient-boosted tree family clusters tightly at 35% improvement, while the MLP achieves the best result (−36.6%).

Table 5.2: Per-dataset MAE on native rating scale (XGBoost, 5-fold CV).

Dataset	Scale	MAE	Spearman ρ
SCUT-FBP5500	1–5	0.38	0.76
LiveBeauty	1–5	0.32	0.71
MEBeauty	1–10	0.74	0.65

5.2 Decision Tree Analysis

As a symbolic, fully interpretable model we train a Decision Tree (max depth 7, 124 leaves). Figure 5.1 shows the top three split levels. The root splits on `eye_area_ratio`, separating

¹Percentage reduction in MAE computed in z-score space (baseline MAE = 0.858).

below-average ($\bar{z} = -0.33$) from above-average ($\bar{z} = +0.38$) faces. Subsequent levels use `eye_width_ratio`, `nose_width_ratio`, and `gonial_angle`, confirming the SHAP findings. Despite lower accuracy ($\rho=0.54$ vs. 0.70 for XGBoost), the tree yields directly readable rules mapping facial proportions to attractiveness.

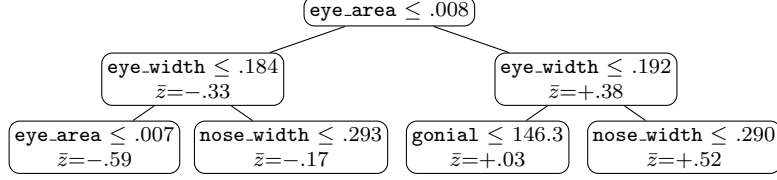


Figure 5.1: Top three levels of the Decision Tree. Left children take the $\leq$ branch.

5.3 Cross-Dataset Generalization

Table 5.3: Cross-dataset generalization (train on one dataset, test on another). MAE on test dataset’s native scale.

Experiment	Test Scale	MAE	Spearman ρ	Test N
SCUT $\rightarrow$ MEBeauty	1–10	1.13	0.29	2,370
SCUT $\rightarrow$ LiveBeauty	1–5	0.44	0.50	10,000
MEBeauty $\rightarrow$ SCUT	1–5	0.53	0.37	5,500
MEBeauty $\rightarrow$ LiveBeauty	1–5	0.43	0.51	10,000
LiveBeauty $\rightarrow$ MEBeauty	1–10	1.14	0.43	2,370
LiveBeauty $\rightarrow$ SCUT	1–5	0.50	0.52	5,500

The worst generalization occurs for **SCUT** $\rightarrow$ **MEBeauty** ($\rho=0.29$), confirming that a model trained on only Asian and Caucasian faces fails when tested on a diverse population (Black, Hispanic, Indian, Middle Eastern subjects). This validates research question (1). Transfer between datasets sharing an ethnic group works moderately: SCUT $\leftrightarrow$ LiveBeauty ($\rho=0.50$ –0.52) both contain majority-Asian faces, suggesting shared geometric patterns within an ethnic group across different rater populations.

5.4 Fairness Analysis: Multi-Model Bias Comparison

To understand whether model choice affects fairness, we compute per-ethnicity MAE for multiple model families and report the **absolute fairness gap** (maximum group MAE minus minimum group MAE) as well as the per-group deviation from the overall MAE.

Table 5.4: Per-ethnicity MAE (z-score) across model families (holdout test set). Gap = max – min group MAE.

Model	Asian	Cauc.	Black	Hisp.	Mid-E.	Indian	Gap
Ridge	0.587	0.616	0.582	0.762	0.743	1.038	0.456
Decision Tree	0.648	0.648	0.689	0.840	0.716	1.007	0.359
Random Forest	0.605	0.635	0.607	0.780	0.686	1.020	0.415
XGBoost	0.552	0.601	0.616	0.717	0.718	0.876	0.324
MLP	0.531	0.575	0.593	0.698	0.703	0.851	0.320

Key observations: (1) The absolute fairness gap ranges from 0.320 (MLP) to 0.456 (Ridge), showing that more expressive models reduce but do not eliminate ethnic disparities. (2) The Indian subgroup consistently has the highest MAE across all models, confirming that the bias is driven by *data scarcity* (N=156) rather than any particular model’s inductive bias. (3) The gap between Asian and Indian MAE is approximately 0.3–0.45 z-score units regardless of model family, suggesting that the fairness problem is fundamentally a data problem: no model architecture can compensate for having 92× fewer training examples for Indian faces than Asian faces.

The gender gap is smaller: male MAE = 0.54, female MAE = 0.61 (absolute gap: 0.07).

5.5 Feature Importance and SHAP Analysis

Table 5.5: Top-5 features by mean absolute SHAP value, per ethnicity.

Ethnicity	Top-5 Features (ranked)	Top SHAP
Asian	nose_width, eye_width, mouth_chin, eye_area, lip_fullness	0.264
Black	nose_width, eye_width, mouth_chin, eye_area, lip_fullness	0.677
Caucasian	nose_width, eye_width, mouth_chin, eye_area, brow_arch	0.263
Hispanic	nose_width, eye_width, mouth_chin, eye_area, brow_arch	0.319
Indian	nose_width, eye_width, mouth_chin, eye_area, jaw_width	0.330
Mid-Eastern	nose_width, eye_width, mouth_chin, eye_area, lip_fullness	0.318

The top-4 features (**nose width ratio**, **eye width ratio**, **mouth-chin ratio**, and **eye area ratio**) are identical across all ethnic groups, as determined by TreeExplainer SHAP values [9]. Only the 5th feature varies: lip fullness for Asian, Black, and Mid-Eastern faces; brow arch height for Caucasian and Hispanic; jaw width for Indian. The magnitude of influence differs: for Black faces, nose width ratio has 2.6× the SHAP importance (0.677) compared to Asian faces (0.264). This suggests the model relies more heavily on nose geometry to distinguish attractiveness within the Black subgroup, possibly because nose proportions show higher within-group variance for this ethnicity.

Notably, **canthal tilt**, despite its popularity in online beauty discourse, ranks 17th–23rd out of 39 features across all groups, suggesting it has minimal predictive value when other geometric features are available. Symmetry-based features (facial symmetry, eye

symmetry) show Pearson correlation with attractiveness below $r = 0.04$, likely reflecting MediaPipe’s design: landmark detection places symmetric landmarks by construction, so symmetry features measure detection consistency rather than actual facial asymmetry. Eye asymmetry (difference in left-right eye height) conversely shows mild signal at $r = -0.11$.

5.6 Comparison to Benchmarks

5.6.1 vs. Most-Upvoted Kaggle Notebook

The most-upvoted Kaggle notebook [10] for SCUT-FBP5500 trains a 4-layer CNN from scratch on raw 128×128 pixel images (10 epochs, no pretraining, no augmentation), achieving $\text{RMSE} \approx 0.51$ on a single 80/20 split. Our XGBoost model trained on SCUT-only data achieves $\text{RMSE} \approx 0.44$ under 5-fold stratified CV, outperforming it by approximately 15% despite using only 30 interpretable geometric measurements rather than raw pixels.

5.6.2 vs. Academic SOTA (FPEM)

The Facial Physical Expectation Model (FPEM) [11] represents the current academic state-of-the-art, combining geometric and appearance features with a deep learning architecture. It benchmarks on all three of our datasets:

Table 5.6: Comparison with FPEM (academic SOTA) by Spearman ρ .

Dataset	FPEM	Ours (XGBoost)	Gap
SCUT-FBP5500	0.931	0.700	−0.231
LiveBeauty	0.925	0.706	−0.219
MEBeauty	0.787	0.648	−0.139
Cross-dataset (LB→ME)	0.640	0.418	−0.222

The gap is smallest on MEBeauty (the most ethnically diverse dataset), suggesting that appearance cues help more on homogeneous data where geometric variation is lower. FPEM’s cross-dataset transfer ($\rho=0.64$) also degrades substantially from its within-dataset performance, confirming that cross-cultural generalization remains an open challenge regardless of approach.

5.7 Limitations

- **Data imbalance:** With 80% Asian faces, results for minority groups have wider confidence intervals and the absolute fairness gap (0.32 z-score) reflects insufficient representation.

- **Annotation bias:** Attractiveness ratings reflect rater demographics (predominantly East Asian for LiveBeauty, mixed for MEBeauty), which we cannot fully decouple from subject ethnicity.
- **Texture exclusion:** By design, our model ignores skin texture, color, hair, and grooming. These features undoubtedly influence attractiveness judgments and explain most of the gap to CNN methods.
- **MediaPipe symmetry artifact:** The landmark detector places left-right landmarks symmetrically by construction. Symmetry features therefore measure detection consistency rather than actual facial asymmetry, explaining their near-zero predictive power.
- **Regression-to-the-mean:** All models show std ratios below 1.0 (0.63–0.85), indicating compressed prediction ranges.
- **LiveBeauty distortion:** The near-binary score distribution inflates aggregate metrics without adding genuine regression signal.

6 Conclusion

We demonstrated that 30 interpretable geometric features from facial landmarks predict human attractiveness ratings with meaningful accuracy (MAE 0.38 on SCUT 1–5, 0.74 on MEBeauty 1–10) when trained on a diverse corpus of 17,870 faces across 3 datasets and 6 ethnic groups. Compared to the academic SOTA (FPEM, $\rho=0.93$ on SCUT), our geometry-only approach achieves $\rho \approx 0.70$; the gap reflects texture cues (skin, hair, grooming) that landmarks cannot capture.

Cross-dataset experiments confirm that models trained on ethnically homogeneous data (SCUT: Asian/Caucasian only) generalize poorly to diverse populations ($\rho = 0.29$ on MEBeauty), while combining heterogeneous datasets improves both performance and cross-ethnic fairness. Multi-model fairness analysis reveals an absolute gap of 0.32–0.46 z-score units between best- and worst-served groups, consistent across all model families, indicating the disparity is driven by data scarcity rather than architecture.

The key geometric predictors (nose width, eye width, mouth-chin proportions) are consistent across ethnicities, but their magnitudes vary ($2.6\times$ higher SHAP for nose width in Black vs. Asian faces), suggesting beauty perception has both universal and culture-specific components.

Future work should: (1) acquire balanced samples across ethnic groups; (2) integrate rater demographics to disentangle rater-from-ratee bias; (3) combine geometric features with texture embeddings (e.g., ArcFace) to close the CNN gap; and (4) evaluate fairness-aware loss functions.

Bibliography

- [1] Y. Kartynnik, A. Ablavatski, I. Grishchenko, and M. Grundmann, “Real-time facial surface geometry from monocular video on mobile GPUs,” *arXiv preprint arXiv:1907.06724*, 2019.
- [2] L. Liang, L. Lin, L. Jin, D. Xie, and M. Li, “SCUT-FBP5500: A diverse benchmark dataset for multi-paradigm facial beauty prediction,” in *Proceedings of the 24th International Conference on Pattern Recognition (ICPR)*. IEEE, 2018, pp. 1598–1603.
- [3] I. Lebedeva, Y. Guo, and F. Ying, “MEBeauty: a multi-ethnic facial beauty dataset in-the-wild,” *Neural Computing and Applications*, pp. 1–15, 2021.
- [4] F. X. Yu and Z. Liu, “LiveBeauty: A large-scale public facial beauty dataset from live streaming,” *arXiv preprint arXiv:2401.08649*, 2024.
- [5] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [6] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [7] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, “CatBoost: unbiased boosting with categorical features,” *Advances in Neural Information Processing Systems*, vol. 31, 2018.
- [8] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, “Optuna: A next-generation hyperparameter optimization framework,” in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2019, pp. 2623–2631.
- [9] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 4765–4774.
- [10] jhwoo06, “Facial beauty rater,” <https://www.kaggle.com/code/jhwoo06/facial-beauty-rater-hai>, 2024, kaggle notebook.
- [11] F. Li, Y. Zhang, X. Yu, and Z. Liu, “Facial attractiveness prediction and analysis based on facial physical expectation model,” *arXiv preprint arXiv:2501.02509*, 2025.

Ehrenwörtliche Erklärung

Ich versichere, dass ich die beiliegende Bachelor-, Master-, Seminar-, oder Projektarbeit ohne Hilfe Dritter und ohne Benutzung anderer als der angegebenen Quellen und in der untenstehenden Tabelle angegebenen Hilfsmittel angefertigt und die den benutzten Quellen wörtlich oder inhaltlich entnommenen Stellen als solche kenntlich gemacht habe. Diese Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

Declaration of Used AI Tools

Tool	Purpose	Where?	Useful?
claude	code review	Python scripts	+
DeepL	translation	throughout	+
Quillbot	grammar check	throughout	+

Unterschrift

Mannheim, den 17. Mai 2026